

Module 6: Metabolism — Keys to Success

Learning Objectives

1. **Energy Rules:** Apply the Laws of Thermodynamics to life (Energy cannot be created/destroyed; disorder always increases).
 2. **ATP:** Understand how ATP serves as the "energy currency" for cell work.
 3. **Enzymes:** Explain how enzymes speed up reactions like a catalyst or a key in a lock.
 4. **Factors:** List what affects enzyme speed (Temperature, pH, Inhibitors).
 5. **Big Picture:** Connect metabolism to photosynthesis (building up) and cellular respiration (breaking down).
-

Key Concepts & Analogies

1. The Energy Laws (Thermodynamics)

- **First Law:** You can't get something for nothing. Energy is never created, just changed (e.g., from light to food).
- **Second Law:** The universe prefers messiness (Entropy). It takes energy to keep your room clean (maintaining order), just like it takes energy for a cell to stay alive.

2. ATP: The "Cash" of the Cell

- **Food (Glucose):** This is like a gold bar. Valuable, but you can't use it in a vending machine.
- **ATP:** This is like a \$1 coin. The cell breaks down the gold bar (glucose) into many small ATP coins to "pay" for daily work like moving muscles or building proteins.
- **The Cycle:** ATP is used up and becomes ADP. Food energy recharges ADP back into ATP.

3. Enzymes: The Workers

- **What they do:** Speed up chemical reactions that would otherwise be too slow.

- **How they work:**
- **Lock and Key:** An enzyme has a specific shape (active site). Only a specific target (substrate) fits in.
- **Activation Energy:** Enzymes lower the "hump" of energy needed to start a reaction.
- **Reusable:** An enzyme isn't used up. It does its job and moves to the next substrate.

4. What Affects Enzymes?

- **Temperature:** Cold slows them down. Heat speeds them up, but *too much* heat melts/unravels them (**denaturation**). A cooked egg cannot turn back into a raw egg.
- **pH:** Enzymes have a favorite environment (stomach enzymes like acid; intestinal enzymes don't).
- **Inhibitors:** "Stop signals" that block the active site or change the enzyme's shape.

5. Oxidation & Reduction (The "OIL RIG" Mnemonic)

- **Oxidation Is Loss** (of electrons/energy).
 - **Reduction Is Gain** (of electrons/energy).
 - In biology, electrons often travel with a Hydrogen atom. So, if a molecule **gains H**, it is being **reduced** (gaining energy).
-

Essential Vocabulary

Term	Simple Definition
Metabolism	All the chemical reactions happening inside a cell.
Anabolic	Building up. Requires energy. (e.g., Photosynthesis builds sugar).
Catabolic	Breaking down. Releases energy. (e.g., Digestion breaks down food).
Kinetic Energy	Energy of motion (ball rolling, muscle moving).
Potential Energy	Stored energy (ball at top of hill, chemical bonds in food).
Activation Energy	The "push" needed to start a chemical reaction.
Active Site	The "pocket" on an enzyme where the reaction happens.
Denaturation	When a protein loses its shape and stops working (usually due to heat).